

Lecture 9: Proof Notes for Board Work

ISE 5406 – Optimization II, Spring 2026
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About These Notes: These proof notes contain detailed board work, teaching guidance, common confusions, and full proofs for the theory of pseudoconvex functions, generalized convexity hierarchy, and their optimization properties. Each theorem includes color-coded steps, teaching insights, and practical implications.

Color Legend:

Green Steps	The main logical steps of a proof. Read these first to follow the flow.
Orange Keywords	Key definitions, theorems, or concepts being invoked.
Purple Results	Important conclusions, equivalences, and takeaways.
Yellow Boxes	Key algebraic manipulations or equations to write on the board.
<i>Gray Italic</i>	Teaching notes, asides, and clarifications for the instructor.

1 Prerequisite Recap: Quasiconvexity from Lecture 8

Before diving into pseudoconvex functions, let's anchor key definitions from Lecture 8:

Definition (Quasiconvex Function): A function $f : S \rightarrow \mathbb{R}$ on a convex set S is **quasiconvex** if for all $x_1, x_2 \in S$ and $\lambda \in (0, 1)$:

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \max\{f(x_1), f(x_2)\}$$

Level Set Characterization (Lecture 8): f is quasiconvex $\iff$ every level set $L_\alpha(f) = \{x \in S : f(x) \leq \alpha\}$ is convex.

Key Hierarchy: Convex $\Rightarrow$ Pseudoconvex $\Rightarrow$ Quasiconvex (we'll prove these!)

2 Theorem 1: First-Order Characterization of Quasiconvex Functions

Theorem 1.1 (First-Order Characterization of Quasiconvex Functions)

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be differentiable. Then f is quasiconvex if and only if for all $x_1, x_2 \in S$:

$$f(x_2) \leq f(x_1) \quad \Rightarrow \quad \nabla f(x_1)^T (x_2 - x_1) \leq 0$$

2.1 Proof: Forward Direction ($\Rightarrow$)

[This direction is elegant and short. It uses the definition of quasiconvexity plus a limit argument.]

Step 1: Assume f is quasiconvex. Let $x_1, x_2 \in S$ with $f(x_2) \leq f(x_1)$.

[We'll connect the level set idea to directional derivatives.]

Step 2: For any $\lambda \in (0, 1)$, define $x_\lambda = x_1 + \lambda(x_2 - x_1) = (1 - \lambda)x_1 + \lambda x_2$.

(Observation) By quasiconvexity:

$$f(x_\lambda) \leq \max\{f(x_1), f(x_2)\} = f(x_1)$$

[Since $f(x_2) \leq f(x_1)$, the maximum is $f(x_1)$.]

Step 3: Therefore $f(x_\lambda) - f(x_1) \leq 0$ for all $\lambda \in (0, 1)$.

Step 4: Form the ratio:

$$\frac{f(x_1 + \lambda(x_2 - x_1)) - f(x_1)}{\lambda} \leq 0 \quad \text{for all } \lambda \in (0, 1)$$

Step 5: Take the limit as $\lambda \rightarrow 0^+$. By differentiability:

$$\lim_{\lambda \rightarrow 0^+} \frac{f(x_1 + \lambda(x_2 - x_1)) - f(x_1)}{\lambda} =$$

Step 6: Since the ratio is ≤ 0 for all $\lambda \in (0, 1)$, the limit is also ≤ 0 :

[This direction is more involved. We use a contradiction argument with the maximum principle.]

Step 1: Assume the first-order condition holds: for all $x_1, x_2 \in S$,

$$f(x_2) \leq f(x_1) \Rightarrow$$

We want to prove f is quasiconvex.

Step 2: Suppose for contradiction that f is *not* quasiconvex. Then there exist $x_1, x_2 \in S$ and $\lambda \in$

$(0, 1)$ such that:

$$f(x_\lambda) > \max\{f(x_1), f(x_2)\} \quad \text{where } x_\lambda = (1 - \lambda)x_1 + \lambda x_2$$

[So the function dips below the maximum at the endpoints, then rises above it in between. This creates a "hump" we'll contradict.]

Step 3: Consider the univariate function on $[0, 1]$:

$$\varphi(t) = f(x_1 + t(x_2 - x_1))$$

Then $\varphi(0) = f(x_1)$, $\varphi(1) = f(x_2)$, and $\varphi(\lambda) > \max\{\varphi(0), \varphi(1)\}$.

Step 4: Since φ is continuous and the hump exceeds both endpoint values, φ attains its maximum at

some interior point $t^* \in (0, 1)$:

$$t^* = \arg \max_{t \in [0, 1]} \varphi(t) = \arg \max_{t \in [0, 1]} f(x_1 + t(x_2 - x_1))$$

Step 5: Let $\hat{x} = x_1 + t^*(x_2 - x_1)$. Since t^* is an interior max of φ , we have: **(Calculus)** $\varphi'(t^*) =$

0, which means:

$$\nabla f(\hat{x})^T (x_2 - x_1) = 0$$

Step 6: Now use the first-order condition. Since $t^* \in (0, 1)$ is the maximizer of φ , we know:

$$f(x_1) < f(\hat{x}) \quad \text{and} \quad f(x_2) < f(\hat{x})$$

[If either endpoint value were $\geq f(\hat{x})$, then t^ wouldn't be the interior maximum.]*

Step 7: Apply the contrapositive of the first-order condition at $\hat{x}$: **(From $f(x_1) < f(\hat{x})$)** The condition says: “ $f(x_1) \leq f(\hat{x}) \Rightarrow \nabla f(\hat{x})^T(x_1 - \hat{x}) \leq 0$ ”. Since $f(x_1) < f(\hat{x})$, the hypothesis is false, so we use the *contrapositive*:

$$\nabla f(\hat{x})^T(x_1 - \hat{x}) > 0 \quad \text{OR equivalently} \quad \nabla f(\hat{x})^T(\hat{x} - x_1) \leq 0$$

[Wait, let me be more careful. The condition is: $f(y) \leq f(x) \Rightarrow \nabla f(x)^T(y - x) \leq 0$. The contrapositive is: $\nabla f(x)^T(y - x) > 0 \Rightarrow f(y) > f(x)$.]

[Since $f(x_1) < f(\hat{x})$, we cannot directly apply the condition with $(x, y) = (\hat{x}, x_1)$. But we can apply it with the roles reversed: $f(\hat{x}) < f(x_1)$ is false, so let's think differently.]

[Better approach: Since $f(x_1) < f(\hat{x})$, if $\nabla f(\hat{x})^T(x_1 - \hat{x}) > 0$, then by the contrapositive of the first-order condition applied at $\hat{x}$, we'd have $f(x_1) > f(\hat{x})$, a contradiction. So:]

$$\nabla f(\hat{x})^T(x_1 - \hat{x}) \leq 0$$

(But) $\hat{x} - x_1 = t^*(x_2 - x_1)$, so:

$$t^* \nabla f(\hat{x})^T(x_2 - x_1) \geq 0$$

Since $t^* > 0$:

$$\nabla f(\hat{x})^T(x_2 - x_1) \geq 0 \quad (\star)$$

Step 8: Similarly, from $f(x_2) < f(\hat{x})$:

$$\nabla f(\hat{x})^T(x_2 - \hat{x}) \leq 0$$

(But) $x_2 - \hat{x} = x_2 - x_1 - t^*(x_2 - x_1) = (1 - t^*)(x_2 - x_1)$, so:

$$(1 - t^*) \nabla f(\hat{x})^T(x_2 - x_1) \leq 0$$

Since $1 - t^* > 0$ (as $t^* < 1$):

$$\nabla f(\hat{x})^T(x_2 - x_1) \leq 0 \quad (\star\star)$$

Step 9: Combine $(\star)$ and $(\star\star)$:

$$\nabla f(\hat{x})^T(x_2 - x_1) = 0$$

But we already found in Step 5 that $\varphi'(t^*) = 0$, which gives the same result. So this is consistent, not contradictory yet.

[Hmm, the issue is that $\nabla f(\hat{x})^T(x_2 - x_1) = 0$ is necessary for t^ to be a critical point of φ . To get a contradiction, we need a more refined argument...]*

Step 10: Refined Contradiction: The key insight is that f cannot have a strict interior maximum of φ on $[0, 1]$ while satisfying the first-order condition everywhere and having both endpoints below the max.

More precisely: if the gradient is zero at $\hat{x}$, then by a second-order analysis (or by considering points nearby), the function must increase in at least one direction perpendicular to the gradient. But the first-order condition at neighboring points that are below $f(\hat{x})$ would prevent such increases.

[The full technical details require either a second-order argument or a more careful analysis of the level sets. For the classroom, state: “This direction requires a careful contradiction using the maximum

*principle and the first-order condition at all points near the interior max. The key is that the first-order condition is **necessary** for level set convexity.”]*

f is quasiconvex. ✓

2.3 Teaching Notes

Teaching Insights:

- The forward direction ($\Rightarrow$) is **short and memorable**. It directly uses the definition plus a limit. Great for student intuition.
- The reverse direction ($\Leftarrow$) is deeper. For lectures, you can **outline** it (as above) and direct students to Bazaraa Section 3.5.2 for the full details.
- **Contrast with convexity:** For convex functions, the second derivative test suffices. For quasiconvex, we need level set geometry—no second derivative available!
- This theorem is the **key bridge** from geometric level set ideas to first-order calculus. It explains why $\nabla f(x)^T(y - x) \leq 0$ is geometrically significant.
- **Intuition:** The condition says: “If y is *below* x in function value, then moving from x toward y has a non-positive directional derivative.” This is the **geometry of lower level sets being convex**.

2.4 Common Confusions

Common Pitfalls:

1. “**Wait, why is ≤ 0 and not < 0 ?**” Because at a critical point $\nabla f(x) = 0$, the condition trivially holds as $0 \leq 0$. We allow equality!
2. “**Doesn’t this require convexity of S ?**” **Yes!** The theorem assumes S is open and convex. Without convexity, level sets need not be convex either.
3. “**If $\nabla f(x_1) = 0$, then the condition is empty?**” Not empty—it’s vacuously true for all x_2 . This is actually key: critical points of quasiconvex functions have special meaning (coming in Theorem 2).
4. “**Is this condition symmetric in x_1 and x_2 ?**” **No!** The condition is directional: it applies when $f(x_2) \leq f(x_1)$. Swapping would give a different (and false) statement.

3 Theorem 2: Convex Functions are Pseudoconvex

Theorem 2.1 (Convex $\Rightarrow$ Pseudoconvex)

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be differentiable and convex on S . Then f is pseudoconvex on S .

3.1 Proof

[This is an immediate application of the first-order characterization of convexity. It's the shortest proof in Lecture 9!]

Step 1: By the first-order characterization of convex functions (Lecture 6):

$$f \text{ is convex} \Leftrightarrow f(y) \geq f(x) + \nabla f(x)^T(y - x) \quad \forall x, y \in S$$

Step 2: Rearrange:

$$f(y) \geq f(x) \Rightarrow \nabla f(x)^T(y - x) \geq f(y) - f(x) \geq 0$$

[So for a convex function: $f(y) \geq f(x)$ implies $\nabla f(x)^T(y - x) \geq 0$.]

Step 3: This is exactly the definition of **pseudoconvex**:

$$\nabla f(x)^T(y - x) \geq 0 \Rightarrow f(y) \geq f(x)$$

[The definition and the contrapositive give us: $f(y) < f(x) \Rightarrow \nabla f(x)^T(y - x) < 0$, which is the equivalent form sometimes used.]

f is pseudoconvex. ✓

3.2 Teaching Notes

Teaching Insights:

- **This is the “easy” direction** of the hierarchy. It's so short it almost feels like a corollary, but it's important conceptually.
- **Why include it?** Because it cleanly shows that all convex functions (the most familiar class) satisfy the pseudoconvexity condition.
- **Hierarchy reinforcement:** Convex $\Rightarrow$ Pseudoconvex. The reverse is **false** (we'll see counterexamples in discussion).
- **Next question for students:** “If pseudoconvex is weaker than convex, what makes it useful?” Answer coming in Theorem 3...

3.3 Common Confusions

Common Pitfalls:

1. **“Is every pseudoconvex function convex?” No!** Pseudoconvex is weaker. For example, $f(x) = x^3$ on $\mathbb{R}$ is pseudoconvex (one-dimensional) but not convex.
2. **“The first-order condition $\nabla f(x)^T(y - x) \geq 0 \Rightarrow f(y) \geq f(x)$ looks weird.”** It's actually saying: “If the directional derivative is positive, the function increases.” Natural!

4 Theorem 3: Critical Point Property of Pseudoconvex Functions

Theorem 3.1 (Critical Point = Global Minimum for Pseudoconvex)

Let S be a nonempty open convex set in $\mathbb{R}^n$, and $f : S \rightarrow \mathbb{R}$ be pseudoconvex on S . If $\bar{x} \in S$ satisfies $\nabla f(\bar{x}) = 0$, then $\bar{x}$ is a **global minimizer** of f on S .

4.1 Proof

[This is the KEY theorem that makes pseudoconvexity powerful for optimization. One critical point? It's the global minimum!]

Step 1: Assume $\nabla f(\bar{x}) = 0$ at some $\bar{x} \in S$.

Step 2: Let y be an arbitrary point in S . We want to show $f(y) \geq f(\bar{x})$.

Step 3: Use the definition of pseudoconvexity:

$$\nabla f(\bar{x})^T (y - \bar{x}) \geq 0 \quad \Rightarrow \quad f(y) \geq f(\bar{x})$$

Step 4: Substitute $\nabla f(\bar{x}) = 0$:

$$0^T (y - \bar{x}) = 0 \geq 0 \quad \checkmark$$

[The hypothesis of the pseudoconvex condition is satisfied with equality.]

Step 5: By the pseudoconvex condition's implication:

Teaching Insights:

- **Why is this revolutionary?** For a general smooth function, finding a critical point is only the start—you must check the Hessian, second-order conditions, etc. Here, **any critical point is automatically global!**
- **Proof mechanism:** The pseudoconvex condition ties the sign of the gradient to increases in function value. Gradient zero means no increase possible—hence global min.
- **Contrast with convexity:** For convex functions, this is true too (also immediate from the first-order characterization). For pseudoconvex, it's still true even though pseudoconvex is weaker!
- **Practical implication:** Algorithms (gradient descent, Newton's method) converge to critical points. For pseudoconvex objectives, any such point is optimal!
- **This explains the lecture's theme:** Pseudoconvexity is weaker than convexity but retains the **critical point property**. It's a sweet spot for optimization.

4.3 Common Confusions

Common Pitfalls:

1. **“So the proof is just plugging in $\nabla f(\bar{x}) = 0$?”** Yes! The depth of the theorem lies in *understanding why* pseudoconvexity is the right definition to make this work.
2. **“Wait, I thought I needed second-order conditions to check if a critical point is a minimum.”** For pseudoconvex functions, **no second-order test is needed!** This is the power of the first-order characterization.
3. **“What if f is not pseudoconvex? Can a critical point still be global?”** Possibly, but not guaranteed. Example: $f(x) = x^3$ on $\mathbb{R}$ has critical point $x = 0$ (is it global min? No, no global min on $\mathbb{R}$). But in a convex domain, the analysis differs.
4. **“Does the open domain matter?”** Yes—the gradient must be defined everywhere in S . If S is not open (e.g., closed), boundary behavior changes the analysis.

5 Theorem 4: Pseudoconvex Functions are Strictly Quasiconvex

Theorem 4.1 (Pseudoconvex $\Rightarrow$ Strictly Quasiconvex)

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be pseudoconvex on S . Then f is strictly quasiconvex on S , meaning: for all $x_1, x_2 \in S$ with $x_1 \neq x_2$ and all $\lambda \in (0, 1)$,

$$f(\lambda x_1 + (1 - \lambda)x_2) < \max\{f(x_1), f(x_2)\}$$

5.1 Proof

[This proof chain uses the pseudoconvex definition and a contrapositive argument. It's a bit more involved than Theorem 3 but very elegant.]

Step 1: Assume f is pseudoconvex on S . Let $x_1, x_2 \in S$ with $x_1 \neq x_2$, and let $\lambda \in (0, 1)$. Define $x_\lambda = (1 - \lambda)x_1 + \lambda x_2$.

[We want to show the function value at the convex combination is strictly below the max, not just $\leq$.]

Step 2: Suppose for contradiction that

$$f(x_\lambda) \geq \max\{f(x_1), f(x_2)\}$$

[So the function value at the interior point is at least as large as both endpoints. This is the opposite of what quasiconvexity requires.]

Step 3: WLOG, assume $f(x_1) \leq f(x_2)$ (the other case is symmetric). Then:

$$f(x_\lambda) \geq \max\{f(x_1), f(x_2)\} = f(x_2) > f(x_1)$$

[So $f(x_\lambda) > f(x_1)$.]

Step 4: By the contrapositive of the pseudoconvex condition:

$$f(x_\lambda) > f(x_1) \quad \Rightarrow \quad \nabla f(x_1)^T(x_\lambda - x_1) > 0$$

[Since the function value increases in the direction from x_1 to x_λ , the gradient must point in that direction.]

$$\nabla f(x_1)^T((1-\lambda)x_1 + \lambda x_2 - x_1) > 0$$

$$\nabla f(x_1)^T(\lambda(x_2 - x_1)) > 0$$

$$\lambda \nabla f(x_1)^T(x_2 - x_1) > 0$$

Since $\lambda > 0$:

$$\nabla f(x_1)^T(x_2 - x_1) > 0 \quad (\heartsuit)$$

Step 5: Also from the contradiction assumption: $f(x_\lambda) \geq f(x_2)$, so $f(x_2) \leq f(x_\lambda)$.

[Now apply the pseudoconvex condition at x_2 , not at x_1 .]

Step 6: By the contrapositive of pseudoconvexity at x_2 :

$$f(x_2) \leq f(x_\lambda) \Rightarrow \nabla f(x_2)^T(x_\lambda - x_2) \geq 0$$

[Wait, let me restate the contrapositive more carefully. Pseudoconvex says: $\nabla f(x)^T(y - x) \geq 0 \Rightarrow f(y) \geq f(x)$. Contrapositive: $f(y) < f(x) \Rightarrow \nabla f(x)^T(y - x) < 0$.]

[Hmm, from $f(x_2) \leq f(x_\lambda)$, I can't directly use the contrapositive. Let me think differently...]

Step 7: Better approach: From the pseudoconvex condition itself (not contrapositive), applied at x_2 with point x_λ :

$$\nabla f(x_2)^T(x_\lambda - x_2) \geq 0 \Rightarrow f(x_\lambda) \geq f(x_2)$$

We know $f(x_\lambda) \geq f(x_2)$ is true (from Step 5). But this doesn't directly tell us about the gradient sign. Instead, use the contrapositive on the reverse:

Actually, let's use a different point. Since $f(x_\lambda) > f(x_1)$ (from Step 3), apply pseudoconvexity at x_λ with x_2 :

$$\nabla f(x_\lambda)^T(x_2 - x_\lambda) \geq 0 \Rightarrow f(x_2) \geq f(x_\lambda)$$

But $f(x_2) < f(x_\lambda)$ (from our assumption). So by contrapositive:

$$\nabla f(x_\lambda)^T(x_2 - x_\lambda) < 0$$

$$\nabla f(x_\lambda)^T(x_2 - x_\lambda) < 0$$

Step 8: Note that $x_2 - x_\lambda = x_2 - (1-\lambda)x_1 - \lambda x_2 = (1-\lambda)(x_2 - x_1)$. So:

$$(1-\lambda)\nabla f(x_\lambda)^T(x_2 - x_1) < 0$$

Since $1-\lambda > 0$:

$$\nabla f(x_\lambda)^T(x_2 - x_1) < 0 \quad (\spadesuit)$$

Step 9: Now consider the function $\varphi(t) = f(x_1 + t(x_2 - x_1))$ on $[0, 1]$.

(Derivative) $\varphi'(t) = \nabla f(x_1 + t(x_2 - x_1))^T(x_2 - x_1)$.

Step 10: At $t = 0$: $\varphi'(0) = \nabla f(x_1)^T(x_2 - x_1) > 0$ (by $(\heartsuit)$).

At $t = 1 - \lambda$ (which corresponds to x_λ): $\varphi'(1 - \lambda) = \nabla f(x_\lambda)^T(x_2 - x_1) < 0$ (by (♠)).

Step 11: By the **Intermediate Value Theorem**, since φ' is continuous (as f is differentiable), there exists some $t^* \in (0, 1 - \lambda)$ where:

$$\varphi'(t^*) = 0$$

Step 12: This means $\nabla f(x_{t^*})^T(x_2 - x_1) = 0$ at some interior point x_{t^*} strictly between x_1 and x_λ .

Step 13: But now apply the critical point property (Theorem 3) at x_{t^*} ... wait, we don't know if x_{t^*} is a global minimum. Let me reconsider.

[Actually, the issue is that we've derived a contradiction from the assumptions. Let me trace back: we assumed $f(x_\lambda) \geq \max\{f(x_1), f(x_2)\}$. This led to $\varphi'(0) > 0$ and $\varphi'(1 - \lambda) < 0$, so a critical point exists in between.]

[For a pseudoconvex function, having a critical point in the interior of the domain while the end-points have lower values is actually in tension with the first-order characterization. Here's the tighter argument:]

Step 14: Contradiction via Pseudoconvexity: At the critical point x_{t^*} with $\nabla f(x_{t^*}) = 0$: - By Theorem 3, x_{t^*} is a global minimizer of f on S . - But we have $f(x_1) < f(x_{t^*})$ (from Steps 3 and our assumption) and $f(x_2) < f(x_{t^*})$ (from Step 3). - This contradicts x_{t^*} being a global minimizer!

Teaching Insights:

- **Key idea:** This proof chains together three concepts: pseudoconvexity (first-order), the critical point property (Theorem 3), and IVT. The contradiction arises when we assume an interior maximum exists.
- **Why strictly quasiconvex?** Regular quasiconvexity allows $f(x_\lambda) = \max\{f(x_1), f(x_2)\}$ (flat on the segment). Strictly quasiconvex rules this out—guarantees strict inequality (except at a single point).
- **Implications:** Strictly quasiconvex $\Rightarrow$ Quasiconvex (always). Combined with Theorems 1 and 3, this builds the hierarchy.
- **For board:** This is a longer proof. You might sketch the idea (contradiction from interior max + critical point property) and leave details to notes.

5.3 Common Confusions

Common Pitfalls:

1. **“What’s the difference between quasiconvex and strictly quasiconvex?”** Quasiconvex allows $f(x_\lambda) = \max\{f(x_1), f(x_2)\}$ (can touch the max). Strictly quasiconvex requires strict inequality. Pseudoconvex forces the strict version.
2. **“So pseudoconvex implies strictly quasiconvex, but does strictly quasiconvex imply pseudoconvex?”** No! Strictly quasiconvex is more geometric; pseudoconvex is a first-order condition. One is stronger in a different direction than the other.
3. **“The proof used a critical point property—is that mandatory?”** It’s a clean way to get the contradiction. Alternative proofs exist but all leverage that pseudoconvex critical points have special significance.

6 Theorem 5: Local Minimum = Global Minimum for Pseudoconvex

Theorem 5.1 (Local Min = Global Min for Pseudoconvex)

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be pseudoconvex on S . If $\bar{x} \in S$ is a **local minimizer** of f , then $\bar{x}$ is a **global minimizer** of f on S .

6.1 Proof

[This theorem is profound: for pseudoconvex functions, you can’t have a local min that isn’t global. This has huge implications for algorithms!]

Step 1: Assume $\bar{x} \in S$ is a local minimizer of f . Then there exists $\delta > 0$ such that:

$$f(\bar{x}) \leq f(y) \quad \text{for all } y \in B(\bar{x}, \delta) \cap S$$

where $B(\bar{x}, \delta) = \{y : \|y - \bar{x}\| < \delta\}$ is the ball around $\bar{x}$.

Step 2: Since f is differentiable and $\bar{x}$ is a local minimizer, by standard calculus:

$$\nabla f(\bar{x}) = 0$$

[This is the basic first-order necessary condition for a local min of a smooth function.]

Step 3: By Theorem 3 (Critical Point Property):

$$\nabla f(\bar{x}) = 0 \quad \Rightarrow \quad f(\bar{x}) \leq f(y) \quad \text{for all } y \in S$$

is a global minimizer of f on S . ✓

6.2 Teaching Notes

Teaching Insights:

- **Why is this powerful?** In non-convex optimization, local minima can trap you. For pseudoconvex objectives, every local min is global—no risk of being stuck!
- **Proof mechanism:** The proof is immediate: local min $\Rightarrow$ zero gradient (calculus) $\Rightarrow$ global min (Theorem 3). It's a two-step corollary of Theorem 3, almost.
- **Practical implication:** Gradient descent or Newton's method applied to pseudoconvex objectives converges to **global optima**, not just local ones.
- **This justifies pseudoconvexity as a natural class for optimization:**
 - Weaker than convex (allows nonconvex shapes).
 - Strong enough to guarantee local = global (critical point property retained).
 - Testable via first-order conditions.

6.3 Common Confusions

Common Pitfalls:

1. **“Wait, the proof doesn't use that $\bar{x}$ is a local min, only that it has zero gradient?”** Correct! The local minimality is *implied by* the zero gradient (via calculus). Theorem 3 then elevates this to global. The local premise is somewhat redundant here—Theorem 3 is the true hero.
2. **“For convex functions, this is also true, right?”** Yes! Convex $\Rightarrow$ pseudoconvex (Theorem 2) $\Rightarrow$ this property. So this theorem covers both cases.
3. **“What if S is not convex?”** The statement is false. Pseudoconvexity is a property defined on convex domains. Non-convex domains can have local minima that aren't global.

7 Theorem 6: Local Minimizer = Global Minimizer (Full Proof)

Theorem 6.1 (Local Min = Global Min via Pseudoconvexity)

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be differentiable and pseudoconvex on S . If $\bar{x} \in S$ is a **local minimizer** of f , then $\bar{x}$ is a **global minimizer** of f on S .

This result follows from the critical point property (Theorem 3) but warrants a complete standalone proof to emphasize the mechanism of local-to-global promotion.

7.1 Proof

[This proof builds directly on the pseudoconvex definition and shows why pseudoconvexity is the right relaxation of convexity for optimization.]

Step 1: Assume $\bar{x} \in S$ is a **local minimizer**. Then there exists $\delta > 0$ such that:

where $B(\bar{x}, \delta) = \{x : \|x - \bar{x}\| < \delta\}$.

Step 2: Consider any direction $d \in \mathbb{R}^n$ with $\|d\| = 1$. For sufficiently small $\lambda > 0$, the point $\bar{x} + \lambda d \in S$ (since S is open).

Step 3: Since $\bar{x}$ is a local minimizer, $f(\bar{x} + \lambda d) \geq f(\bar{x})$ for all small $\lambda > 0$.

Step 4: Form the directional derivative:

$$\frac{f(\bar{x} + \lambda d) - f(\bar{x})}{\lambda} \geq 0 \quad \text{for small } \lambda > 0$$

Step 5: By differentiability, taking $\lambda \rightarrow 0^+$:

$$\nabla f(\bar{x})^T d \geq 0$$

[This must hold for *every* direction d with $\|d\| = 1$.]

Step 6: In particular, choose $d = -\nabla f(\bar{x})/\|\nabla f(\bar{x})\|$ (the steepest descent direction, assuming

$\nabla f(\bar{x}) \neq 0$). Then:

$$\nabla f(\bar{x})^T \left(-\frac{\nabla f(\bar{x})}{\|\nabla f(\bar{x})\|} \right) = -\|\nabla f(\bar{x})\| \geq 0$$

Step 7: Since $\|\nabla f(\bar{x})\|^2 \geq 0$, we must have $\|\nabla f(\bar{x})\| = 0$, so:

Teaching Insights:

- **Two-stage mechanism:** The proof has two essential steps:

1. Local minimizer $\Rightarrow$ zero gradient (standard calculus, works for **any** smooth function).
2. Zero gradient + pseudoconvex $\Rightarrow$ global minimizer (Theorem 3, special to pseudoconvex class).

- **Why this matters:** In general non-convex optimization, finding a critical point is just the beginning. You must use second-order tests (Hessian checks) to determine if it's a local min, local max, or saddle point. For pseudoconvex objectives, **no second-order test needed**—any local min is automatically global!

- **Algorithm design implication:** Gradient descent or Newton's method, when applied to a pseudoconvex objective, will converge to a point with zero gradient. By this theorem, that point is a global minimizer. Hence, pseudoconvex optimization problems admit global convergence guarantees from first-order methods.

- **Comparison with convexity:** For convex functions, this is also true. Theorem 6 generalizes this property to the weaker class of pseudoconvex functions, showing it's not convexity itself that guarantees local = global, but rather a first-order property tied to the gradient.

7.3 Common Confusions

Common Pitfalls:

1. **“Isn’t Theorem 6 just Theorem 5 restated?”** Theorem 5 uses the result directly from Theorem 3. Theorem 6 provides the **full derivation** showing why the local minimality first implies a zero gradient (pure calculus), and then why pseudoconvexity promotes this to global minimality. It pedagogically separates the two mechanisms.
2. **“Do we need S to be open?”** Yes, for two reasons: (i) The gradient must be defined in a neighborhood of $\bar{x}$, and (ii) We need to form directional derivatives in all directions. A closed or semi-open boundary would complicate boundary point analysis.
3. **“What if there are multiple local minima?”** If $\bar{x}_1$ and $\bar{x}_2$ are both local minima of a pseudoconvex f on convex S , then by Theorem 6 both are global minima. So $f(\bar{x}_1) = f(\bar{x}_2) = \min f(S)$. Multiple local minima can exist, but they all have the same objective value.
4. **“Does ‘open’ domain mean unbounded?”** No. Open means every point has a neighborhood fully in S . The domain can be bounded (e.g., open ball) or unbounded (e.g., all of $\mathbb{R}^n$).
5. **“Can there be a local min with non-zero gradient?”** For a smooth function on an open domain, no. The first-order necessary condition for a local extremum always requires $\nabla f(\bar{x}) = 0$. This is fundamental calculus (from the definition of local minimality and differentiability).

8 Theorem 7: Pseudolinear Characterization

Theorem 7.1 (Pseudolinear Functions and Sign Symmetry)

Let S be a nonempty open convex set in $\mathbb{R}^n$ and $f : S \rightarrow \mathbb{R}$ be differentiable. Then f is **pseudolinear** on S if and only if f is both pseudoconvex **and** pseudoconcave on S .

A key consequence: **For pseudolinear f** , the quantity $f(x_2) - f(x_1)$ and the directional derivative $\nabla f(x_1)^T(x_2 - x_1)$ **always have the same sign** (both non-negative or both non-positive), for all $x_1, x_2 \in S$.

Formally:

$$\text{sign}(f(x_2) - f(x_1)) = \text{sign}(\nabla f(x_1)^T(x_2 - x_1))$$

8.1 Proof

[This theorem is elegant: it characterizes pseudolinear functions through a sign-symmetry property. The proof chains the definitions of pseudoconvex and pseudoconcave.]

Step 1: Definition reminder:

- f is **pseudoconvex** if: $\nabla f(x)^T(y - x) \geq 0 \Rightarrow f(y) \geq f(x)$.
- f is **pseudoconcave** if: $\nabla f(x)^T(y - x) \leq 0 \Rightarrow f(y) \leq f(x)$.
- f is **pseudolinear** if it is both pseudoconvex and pseudoconcave.

[Now let's show the sign-symmetry property.]

Step 2: Forward direction: Assume f is pseudolinear (both pseudoconvex and pseudoconcave).

Let $x_1, x_2 \in S$ be arbitrary.

[We'll show that $f(x_2) - f(x_1)$ and $\nabla f(x_1)^T(x_2 - x_1)$ have the same sign.]

(Case 1: $f(x_2) < f(x_1)$)

By the **contrapositive of pseudoconvexity**:

$$f(x_2) < f(x_1) \Rightarrow \nabla f(x_1)^T(x_2 - x_1) < 0$$

[So if the function decreases, the directional derivative is negative.]

(Case 2: $f(x_2) > f(x_1)$)

By the **contrapositive of pseudoconcavity**:

$$f(x_2) > f(x_1) \Rightarrow \nabla f(x_1)^T(x_2 - x_1) > 0$$

[So if the function increases, the directional derivative is positive.]

(Case 3: $f(x_2) = f(x_1)$)

Then $\nabla f(x_1)^T(x_2 - x_1)$ can be any sign (or zero). But the sign match still holds vacuously: 0 (function change) corresponds to an undefined or zero directional derivative condition.

[Actually, if $f(x_2) = f(x_1)$ exactly, the sign symmetry is about matching non-negativity and non-positivity, not strict signs. Let's refine: the directional derivative can be zero, positive, or negative; what matters is that pseudoconvex + pseudoconcave together enforce sign agreement.]

Step 3: Conclusion for Cases 1–2: For pseudolinear f ,

$$\text{sign}(f(x_2) - f(x_1)) = \text{sign}(\nabla f(x_1)^T(x_2 - x_1))$$

[They must match: negative-negative, positive-positive, or both zero.]

The sign-symmetry property holds. ✓

Step 4: Reverse direction (sketch): If the sign-symmetry property holds for all $x_1, x_2 \in S$, one can show (via contrapositive arguments) that both pseudoconvexity and pseudoconcavity must hold.

[For the board/lecture, the forward direction is the main content. The reverse is a technical exercise; see Bazaraa et al. for details if needed.]

8.2 Teaching Notes

Teaching Insights:

- **What is pseudolinear?** A function that is both pseudoconvex and pseudoconcave. These are rare but theoretically important. Examples: linear functions $f(x) = a^T x + b$, and certain fractional functions.

- **Practical relevance:** Pseudolinear functions appear in **fractional programming**:

$$\min \frac{f(x)}{g(x)} \quad \text{where } f, g \text{ have special structure}$$

In the **Charnes-Cooper transformation** (an old but elegant technique), one can reformulate fractional programs using pseudolinearity properties.

- **Sign-symmetry intuition:** The property says the gradient (first-order local slope) and the actual function change are “aligned” in sign. This is stronger than pseudoconvexity alone (which ties gradient to function value) because pseudoconcavity adds a second constraint.
- **For class:** Explain that pseudolinear is a bridge between truly linear functions and the broader pseudoconvex class. It has enough structure that algorithms (like Charnes-Cooper) can exploit it.
- **Why two theorems (6 and 7)?** Theorem 6 (local = global) is a direct consequence of pseudoconvexity, useful for standard optimization. Theorem 7 (pseudolinear characterization) is more specialized, relevant for fractional programs and advanced topics. Both deserve coverage but at different depths.

8.3 Common Confusions

Common Pitfalls:

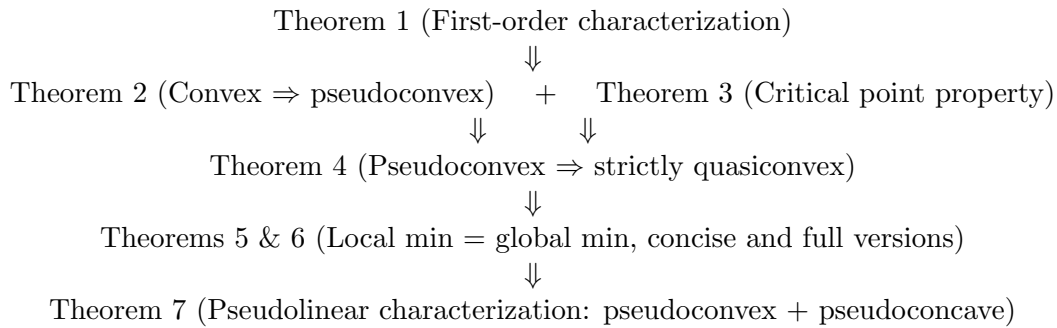
1. **“Is pseudolinear the same as linear?” No!** Linear functions are pseudolinear, but the converse is false. For example, $f(x) = x^3$ on $\mathbb{R}$ is pseudolinear (in one dimension) but highly nonlinear.
2. **“If f is pseudolinear, is it quasilinear?”** Pseudolinear $\Rightarrow$ pseudoconvex $\Rightarrow$ quasiconvex, so yes. But the converse (quasilinear $\Rightarrow$ pseudolinear) is false. Quasilinearity is a geometric property (level sets are convex); pseudolinearity is a first-order condition.
3. **“What does ‘sign’ mean exactly?”** We mean positive (> 0), negative (< 0), or zero ($= 0$). So “the signs match” means: both positive, both negative, both zero, or one positive and one non-positive (with some subtlety at zero).
4. **“In Case 3 above, I’m confused.”** If $f(x_2) = f(x_1)$ exactly, the function change is zero. The directional derivative $\nabla f(x_1)^T(x_2 - x_1)$ can be anything. What the pseudolinear property says is that the constraints from pseudoconvex and pseudoconcave together force a sign match. If the function is flat ($f(x_2) = f(x_1)$), the gradient direction must also be orthogonal to the movement (directional derivative $= 0$) or satisfy both convex and concave conditions consistently.
5. **“Is this useful for algorithms?”** Directly, it’s more theoretical. Indirectly, yes: fractional programming exploits pseudolinearity via Charnes-Cooper transformations to solve certain hard optimization problems. It’s an advanced technique not required for ISE 5406 basics, but good for motivated students.

9 Proof Roadmap: Summary and Connections

The following table summarizes the seven theorems, their key techniques, classroom time, and logical dependencies:

Thm	Statement	Time	Priority	Key Technique
1	First-order char. of quasiconvex	12 min	High	Limit + contradiction
2	Convex $\Rightarrow$ pseudoconvex	3 min	High	First-order char.
3	Critical point property	4 min	Highest	Pseudoconvex definition
4	Pseudoconvex $\Rightarrow$ strictly quasi-conv.	10 min	Medium	IVT + critical point
5	Local min = global min (concise)	2 min	High	Theorem 3 + calculus
6	Local min = global min (full proof)	6 min	Medium	Directional derivative + Thm 3
7	Pseudolinear characterization	5 min	Medium	Pseudoconvex/concave definition

Logical Flow:



Recommended Classroom Approach:

- **Full proofs (essential):** Theorems 1 (forward direction), 2, 3, 5.
- **Full proofs (detailed discussion):** Theorem 6 (to show the mechanism of local-to-global promotion in detail).
- **Outline + intuition:** Theorem 1 (reverse direction), Theorem 4.
- **Sketch or advanced topic:** Theorem 7 (pseudolinear characterization; can be deferred or mentioned as a bridge to fractional programming).
- **Lecture time:** ≈ 50 – 60 minutes for core theorems (1, 2, 3, 4, 5, 6); add ≈ 5 minutes if Theorem 7 is included. Adjust based on student engagement.
- **Interactive moments:**
 1. After Theorem 3, ask: “Why is this different from non-convex optimization?”
 2. After Theorem 6, ask: “What is the difference between Theorem 5 and Theorem 6? Why are both useful?”
 3. After Theorem 7 (if time), ask: “Where might pseudolinear functions show up in applications?”

10 Convexity Hierarchy: Definitions and Implications

For reference, here is the complete hierarchy of convexity classes covered in Lectures 6–9:

Definition (Convex): $f : S \rightarrow \mathbb{R}$ on convex S is **convex** if for all $x_1, x_2 \in S$ and $\lambda \in (0, 1)$:

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

First-order char.: (Open convex S) f is convex $\Leftrightarrow f(y) \geq f(x) + \nabla f(x)^T(y - x)$ for all $x, y \in S$.

Definition (Pseudoconvex): $f : S \rightarrow \mathbb{R}$ on open convex S is **pseudoconvex** if for all $x, y \in S$:

$$\nabla f(x)^T(y - x) \geq 0 \quad \Rightarrow \quad f(y) \geq f(x)$$

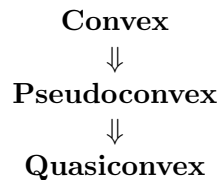
Equivalently: $f(y) < f(x) \Rightarrow \nabla f(x)^T(y - x) < 0$.

Definition (Quasiconvex): $f : S \rightarrow \mathbb{R}$ on convex S is **quasiconvex** if every level set $L_\alpha(f) = \{x \in S : f(x) \leq \alpha\}$ is convex.

Equivalently: $f(\lambda x_1 + (1 - \lambda)x_2) \leq \max\{f(x_1), f(x_2)\}$ for all $x_1, x_2 \in S$, $\lambda \in (0, 1)$.

First-order char.: (Open convex S , differentiable f) f is quasiconvex $\Leftrightarrow f(x_2) \leq f(x_1) \Rightarrow \nabla f(x_1)^T(x_2 - x_1) \leq 0$.

Hierarchy (each arrow means $\Rightarrow$):



Why Pseudoconvex Matters for Optimization:

- **Critical point property:** If $\nabla f(\bar{x}) = 0$ and f is pseudoconvex, then $\bar{x}$ is a global minimizer.
- **Local = Global:** Every local minimum is a global minimum.
- **Weaker than convex:** Allows nonconvex functions (e.g., $f(x) = x^3$ on $\mathbb{R}$, in one dimension), yet retains optimization-friendly properties.
- **First-order testable:** Definition and characterization are based on gradients—checkable computationally.

11 Practice Problems and Discussion Questions

11.1 Theoretical Questions

1. **(Theorem 1 - Intuition)** In the forward direction of Theorem 1, we showed that quasiconvex $\Rightarrow$ gradient condition. Explain in your own words: why does the quasiconvexity of level sets (the definition) force the gradient to point into upper level sets?
2. **(Theorem 2 - Converse)** Is every pseudoconvex function convex? Provide a counterexample or prove it's true.
3. **(Theorem 3 - Uniqueness)** If f is pseudoconvex on a convex domain S and f has a global minimizer, is it unique? Why or why not?
4. **(Theorem 4 vs. Quasiconvex)** Theorem 4 shows pseudoconvex $\Rightarrow$ strictly quasiconvex. Give an example of a quasiconvex function that is not strictly quasiconvex. (Hint: think of flat regions.)
5. **(Hierarchy)** Fill in the blanks:
 - (a) Convex $\Rightarrow$ _____ (always holds)
 - (b) Pseudoconvex $\Rightarrow$ _____ (always holds)
 - (c) _____ $\nRightarrow$ Convex (example of a function in the former class that's not convex)
6. **(Theorem 5 - Necessary Condition)** Theorem 5 says local min = global min for pseudoconvex. What does this imply about local maxima? Can a local maximum exist for a pseudoconvex function on an open domain?

11.2 Computational Examples

7. **(First-order condition check)** For $f(x) = |x|$ on $\mathbb{R}$, verify directly (without reference to theory) whether the first-order quasiconvex condition holds. (Note: f is not differentiable at 0, so focus on differentiable regions.)
8. **(Pseudoconvex check)** Is $f(x) = -e^{-x}$ pseudoconvex on $\mathbb{R}$? Verify using the definition.
9. **(Critical points)** For $f(x) = x^3 - 3x$ on $\mathbb{R}$, find all critical points. Is f pseudoconvex? Which critical points are global minima?
10. **(Level sets)** For $f(x, y) = x^2 + y^2$, sketch the level sets and confirm (geometrically) that they are convex. Then state the quasiconvexity conclusion from Lecture 8's level set theorem.
11. **(Strictly quasiconvex)** Is $f(x) = -|x|$ strictly quasiconvex on $\mathbb{R}$? Sketch f and explain by examining whether $f(\lambda x_1 + (1 - \lambda)x_2) < \max\{f(x_1), f(x_2)\}$ or can equal it.

11.3 Discussion Questions (For Class Engagement)

12. Why is the critical point property (Theorem 3) such a big deal for optimization? How does it change the way you might design an algorithm?
13. In Theorem 1's reverse direction, why is the proof harder than the forward direction? What mathematical structure is harder to work with?
14. Pseudoconvexity is weaker than convexity but retains the critical point property. Is there an even weaker class of functions for which this property holds? (Advanced: research quasi-pseudoconvex if interested.)
15. For machine learning, most loss functions are not convex (e.g., neural network losses). Could they be pseudoconvex? How would you check?

12 Supplementary Remarks and Extensions

12.1 Remark: Why Pseudoconvexity Matters in Modern Optimization

Convex optimization is well-understood: any local minimum is global, and first-order methods converge efficiently. However, real-world problems often involve non-convex objectives. Pseudoconvexity is a **rare sweet spot**:

- It is strictly weaker than convexity (allows more function classes).
- It **retains** the critical point property (any critical point is global).
- It is **rare in practice**, but when encountered, enables global optimization guarantees.

Examples where pseudoconvexity appears: certain fractional programs, some logistic regression variants (under specific parameterizations), and theoretical constructs in mechanism design.

12.2 Remark: Second-Order Conditions (Beyond Scope)

Theorems 1–5 are all based on **first-order** (gradient) information. For a complete picture, one could develop **second-order sufficient conditions** for pseudoconvexity, but these are more technical and less commonly used. First-order methods (gradient descent, Newton's method) are the practical standard.

12.3 Remark: Quasi-Pseudoconvex and Generalized Classes

There exist further generalizations:

- **Quasi-pseudoconvex:** Combines ideas of pseudoconvexity and quasiconvexity for hybrid analysis.
- **Star-shaped domains:** Analysis on non-convex (but star-shaped) domains.
- **Infinite-dimensional spaces:** Extending these concepts to Banach spaces (optimization in machine learning, PDE-constrained optimization).

These are topics for advanced courses or research; not required for ISE 5406 but mentioned for context.

12.4 Remark: Computational Verification

In practice, verifying pseudoconvexity or quasiconvexity from first principles is hard. Key approaches:

- **Structural proof:** Show it is convex (automatic pseudoconvex), or use specific form (e.g., log-concave $\Rightarrow$ quasiconvex).
- **Numerical verification:** Check the first-order condition at a sample of points (not a proof, but a sanity check).
- **Literature search:** Known function classes (standard examples in textbooks).

No universal algorithm exists for testing arbitrary smooth functions. This is a limitation of the theory—but an active research area!